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Navigating the Change: Optimization and Ramp-Up Strategy of a Charging Network for Battery Electric Heavy Trucks

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***Abstract*— This study presents a strategic approach to designing a nationwide public fast-charging infrastructure for battery electric trucks in Germany. First, it optimizes the placement of charging stations at existing rest areas using a Mixed-Integer Quadratic approach, ensuring coverage required by the new EU Alternative Fuels Infrastructure Regulation (AFIR) while optimizing spatial distribution. Second, an optimal ramp-up strategy is calculated to meet intermediate targets and enhance accessibility early on for electric long-haul traffic. The proposed approach integrates Open Street Maps data with traffic flows to identify strategic locations for charging stations, minimizing the total number while maximizing coverage and utility for long-haul freight transport. The results show that the strategic placement of charging stations could reduce initially developed sites by $36\,\%$ compared to previous work while still ensuring $93.8\,\%$ coverage of domestic long-haul traffic in Germany. The proposed ramp-up strategy improves coverage during the phase-in of electric long-haul transport by up to $58.7\,\%$, thereby contributing to the EU's emission reduction targets. This approach facilitates a cost-effective transition to electric freight transport and establishes a scalable model for EU-wide implementation.**

INTRODUCTION

The European Union (EU) CO_2 emission reduction targets require truck manufacturers to reduce vehicle emissions by $45\,\%$ by 2030 and $90\,\%$ by 2040 [1]. As the Diesel technology only improved by $9.5\,\%$ over the past 3 decades [2], the necessary improvements in local emissions can only be achieved through electrification. The most promising technology economically and ecologically are Battery Electric Trucks (BET) [3]. Heavy trucks, despite making up only $7\,\%$ of commercial vehicles, contribute $48\,\%$ to the emissions of road freight transport. Thus the electrification of heavy trucks is the biggest lever in the effort to decarbonize the transport sector. To enable transport that extends beyond the battery range from private charging infrastructure, a public charging network is required. Initially, the build-up of this network is tied to high investment costs and low revenue due to the gradual market introduction of BET. To assure a basic level of service, from 2030 onwards the EU Alternative Fuel Infrastructure Regulation (AFIR) mandates a maximum gap of $60\,\mathrm{km}$ between charging sites along the core Trans-European Transport Network (TEN-T) network and $100\,\mathrm{km}$ along the comprehensive TEN-T network [4]. As an intermediate goal, $50\,\%$ coverage has to be provided from 2028 on.

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This publication sets out to merge these requirements into a holistic build-up strategy: The main goal is to implement an algorithm, that yields an economically viable solution for the electrification of existing rest areas, such that the AFIR requirements are met with a minimal effort and maximal utility. Following this, an optimal ramp-up strategy is crafted to electrify as much traffic flow as possible early on, enabling early implementation of long-haul electric goods transport. Finally, the results of the algorithmic approaches will be interpreted to derive general guidelines for the creation of a charging network for BET.

While this study primarily concentrates on Germany, the methodologies and data sources employed are designed with scalability in mind to be applicable throughout Europe. The main reasons are, that the limited area makes in-depth analysis of the results feasible, and that Germany serves as a hub for European road freight transport. Besides 5 TEN-T corridors transiting the country [5], Germany itself is heavily reliant on road freight transport for domestic traffic. The successful electrification of the German motorways is thus a precondition to electric road freight transport on a European level.

Within this paper we will address the problem of designing an efficient charging network and a ramp-up strategy for it, addressing the complex needs of transitioning towards electric freight transport in a structured manner.

RELATED WORK

Charging networks for electric vehicles in general [6, 7, 8] and trucks in particular [9, 10, 11, 12, 13, 14] are the subject of intense research efforts. In the terms of charging network design, the trade-off between low investment budgets and maximum utility in terms of coverage with charging infrastructure or occupancy is a significant conflict of objectives [14, 15], highlighted by the necessary integration into the existing electricity grid [14, 16]. A further crucial aspect are waiting times at charging stations due to limited site capacities, which are economical disadvantages in commercial vehicles [12, 17]. Each site to be electrified requires an electric grid connection, project management and construction works. The concentration of plugs at sites thus reduces investment costs, yet previous publications plan low numbers of charging points per site [11], even down to a single charging point [14].

Simulations of possible public charging infrastructure for trucks in Germany have been carried out previously [11, 13, 14, 18]. In [11, 13, 14], the problem is approached through time-forward simulation of trucks and queues at

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charging infrastructure. Systems of non-interacting agents on varying charging networks are analyzed in [18]. Yet no study to date provides a comprehensive approach to fulfill the coverage targets set by AFIR on the TEN-T corridors optimally. Especially, start-up and expansion networks are computed independently [11], meaning expansion networks are not augmentations of the startup networks presented. When there is a continuity, only the target states are given [12, 14]. Furthermore, a ramp-up strategy that achieves the required coverage goals, while providing maximal coverage at any stage of expansion is a significant research gap.

Hurtado-Beltran et al. [16] investigate the feasibility of a network of fast-charging stations for electric trucks on the U.S. Interstate Highway System. The main goal is the estimation of road network coverage if fast-charging stations were located at existing truck stop facilities along this system. Utilizing GIS network analysis, the study identifies the coverage of potential infrastructure could provide for long-haul routes. In following scenarios, the proximity of sites to high-voltage electric transmission lines is constrained to assure that a charging site can be supplied with electricity. The study found that truck stop facilities could potentially provide $60\,\%$ to $99.5\,\%$ road network coverage for electric trucks on the Interstate Highway System, depending on the assumed range of vehicles and allowed distance of charging sites to high-voltage transmission lines. In the most restrictive scenario, $40\,\mathrm{km}$ of coverage radius is assumed for each charging site and $1.6\,\mathrm{km}$ of distance to high voltage transmission lines is permitted. To achieve optimal coverage in the restrictive scenario, the electrification of 348 truck stop sites is required. By iterative pruning of the network, the number of sites is reduced to 162, equalling to one station per $594\,\mathrm{km}$ of motorways. In the case of long vehicle ranges, one charging site per $2189\,\mathrm{km}$ of Interstate network suffices.

This article aims to build up on the state of the art and amend it in key points. Firstly, the layout of the charging network is being optimized under multiple constraints: A budget of charging sites ensures an economically viable concentration of charging points per site. Simultaneously, the spatial coverage has to comply with the new EU AFIR regulation criteria. In this context, the inhomogenous coverage requirements on core and comprehensive network are addressed. The candidate locations are constrained to existing service and rest areas. Finally, an incremental strategy for the ramp-up of the charging network is conceived and evaluated. Throughout the publication, we adhere to the principle of open data that is available throughout Europe, to ensure the scalability of the developed method.

METHODOLOGY

The approach is structured into two steps: First, an optimized target network configuration is computed, then ramp-up strategies to incrementally build this target state are evaluated. An overview of the approach and the data sources is provided within the graphical abstract Figure 1.

Network optimization. In order to plan the charging network, an optimization problem is formulated. The objective is, that the full TEN-T core network is covered with maximum permissible gaps, and that charging sites are spaced as uniformly as possible. Furthermore, it is desirable to develop as few sites as possible initially in order to reduce investment costs. In the following paragraph, the problem modeling will be discussed in detail.

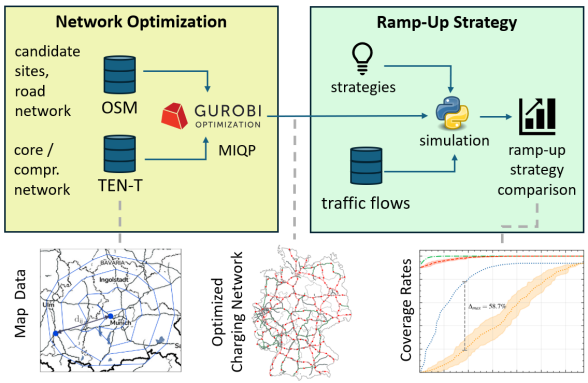


Fig. 1. Graphical abstract of the work presented. This publication relies on open data and a two-step approach: Firstly, the optimal charging network layout fulfilling the given constraints is computed, secondly, the ramp-up strategy is analyzed.

The objective of the optimization is to find a configuration, in which charging sites are distributed evenly throughout the network, to offer regular charging opportunities to passing long-haul transport. This objective is reformulated as a penalty for closely located sites. Let $\underline{\underline{\mathbf{D}}}$ be the $c \times c$ distance matrix among all candidate sites. The Objective Function Matrix $\underline{\underline{\mathbf{Q}}}$ is then computed, by replacing each entry d_{ij} with $q_{ij} = \frac{1}{d_{ij}^2}$, or zero if $i = j$. Binary-valued vector $\underline{\mathbf{x}}$ describes the portfolio of candidate sites to be equipped with charging infrastructure. Then, the term $\underline{\mathbf{x}}^T \underline{\underline{\mathbf{Q}}} \underline{\mathbf{x}}$ sums up all entries q_{ij} where x_i and x_j are both equal to 1. Followingly, the objective function is the sum of distance penalties among candidate sites selected into the portfolio (Equation 2).

$$\underline{\underline{\mathbf{D}}} = \begin{pmatrix} d_{11} & \dots & d_{1c} \\ \vdots & \ddots & \vdots \\ d_{c1} & \dots & d_{cc} \end{pmatrix} \tag{1}$$

$$\begin{aligned} \underline{\mathbf{x}}^T \underline{\underline{\mathbf{Q}}} \underline{\mathbf{x}} &= x_1 x_1 q_{11} + x_1 x_2 q_{12} + x_1 x_3 q_{13} + \dots \\ &= 0 \cdot x_1 x_1 + \frac{x_1 x_2}{d_{12}^2} + \frac{x_1 x_2}{d_{13}^2} + \dots \end{aligned} \tag{2}$$

To formulate the coverage constraints mandated by the AFIR, the distances between individual road segments and candidate sites are considered. The AFIR mandates maximum permissible gaps d_{max} between charging sites. This is remodeled, such that no road segment of the TEN-T core network shall be further than $\frac{d_{max}}{2}$ from the closest charging site. With $\underline{\underline{\mathbf{A}}}$ as the distance matrix of size $s \times c$ between and s road segments and c candidate sites, a coverage constraint of the road network can be derived from $\underline{\underline{\mathbf{A}}}$. The entries of $\underline{\underline{\mathbf{A}}}$ are replaced by zeros, if the distance to the respective charging site is greater than d_{max}, elsewise by ones, to obtain

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coverage matrix $\underline{\underline{\mathbf{A}}}_c$. All road segments lie within $d_{max}/2$ of a charging site, if product $\underline{\underline{\mathbf{A}}}_c \underline{\mathbf{x}}$ is at least 1 in all rows (Equation 3).

$$\underline{\underline{\mathbf{A}}}_c = \begin{pmatrix} f(a_{11}) & \dots & f(a_{1c}) \\ \vdots & \ddots & \vdots \\ f(a_{s1}) & \dots & f(a_{sc}) \end{pmatrix}, f(x) = \begin{cases} 0, & x > \frac{d_{max}}{2} \\ 1, & x \leq \frac{d_{max}}{2} \end{cases}$$
$$\underline{\underline{\mathbf{A}}}_c \underline{\mathbf{x}} \geq \begin{pmatrix} 1 & 1 & \dots \end{pmatrix}^T = \underline{\mathbf{b}}_c \tag{3}$$

To account for inhomogenous coverage constraints on the TEN-T core and comprehensive networks, the matrix is computed twice as $\underline{\underline{\mathbf{A}}}_{core}$ and $\underline{\underline{\mathbf{A}}}_{comp}$ with $d_{max,core}/2$ and $d_{max,comp}/2$ respectively.

Finally, the budget of charging sites to be electrified is also expressed as a vector operation (Equation 4). This budget is initialized with a reasonably high value (1 station per $30\,\mathrm{km}$) and parametrized by iteratively reducing the budget until infeasibility of the problem is reached.

$$\begin{pmatrix} 1 & 1 & \dots \end{pmatrix} \underline{\mathbf{x}} = \underline{\mathbf{c}}^T \underline{\mathbf{x}} \leq Nc \tag{4}$$

To improve convergence, entries in $\underline{\underline{\mathbf{Q}}}$ corresponding to distances greater than $65\,\mathrm{km}$ are set to zero before solving the problem, as these candidate sites lie outside of the permissible gaps in any case and a discontinuity of the objective function at $60\,\mathrm{km}$ is to be avoided.

By concatenating the matrices $\underline{\underline{\mathbf{A}}}_{core}$ and $\underline{\underline{\mathbf{A}}}_{comp}$ and the vector $\underline{\mathbf{c}}^T$ final optimization problem formulation is obtained (Equation 5). It takes the form of a mixed-integer quadratic problem and can be solved using Gurobi Optimizer version 11.0.0 [19] in $49\,\mathrm{s}$ on an Intel® Core™ i7-10850H CPU (Ubuntu 22.04).

$$\min_{\underline{\mathbf{x}}} \quad \underline{\mathbf{x}}^T \underline{\underline{\mathbf{Q}}} \underline{\mathbf{x}} \tag{5}$$
$$\text{subject to} \quad \begin{pmatrix} \underline{\underline{\mathbf{A}}}_{core} \\ \underline{\underline{\mathbf{A}}}_{comp} \\ -\underline{\mathbf{c}}^T \end{pmatrix} \underline{\mathbf{x}} \geq \begin{pmatrix} \underline{\mathbf{b}}_{core} \\ \underline{\mathbf{b}}_{comp} \\ -N_c \end{pmatrix}$$
$$\underline{\mathbf{x}} \in \mathbb{N}^c, x_i \leq 1$$

Ramp-up strategy. In order to meet the intermediate coverage targets of $50\,\%$ in 2028, and to generally provide a good coverage early on, the next paragraph examines strategies to incrementally select candidate sites for the build-up of charging infrastructure. For this purpose, all routes of domestic long-haul traffic in Germany are evaluated. The possible charging stations are projected onto the route, and combined with the simulated energy consumption. With this information, the minimal State of Charge (SOC) during such a journey is determined. The resulting feasibility or infeasibility is then multiplied with the actual annual traffic flow on the respective route. The underlying process of consumption prediction has been previously published [20] and the application to real world road networks was researched in [18]. The traffic flow data is obtained from [21]. For this publication, the traffic flows from the year 2019 are employed, but data for the year 2030 is also available.

In the following, two approaches for a ramp-up are compared: *Greedy* and *random*. The random approach forms the baseline and aims to represent an uncoordinated effort to build the charging network. Incrementally, a random site among the selected optimized sited is picked and added to the network. This process is repeated $n = 10$ times to provide insights on the variance among the performance of these random strategies.

The greedy strategy represents an informed approach aiming for maximum return on investment for charging point operators. The operator has perfect knowledge and decides unilaterally optimally. At each decision point, the electrification of all single candidate sites is simulated and the one site enabling the greatest traffic flow to be electrified is picked. Subsequently, all remaining charging stations are iteratively evaluated again and the best station is added to the list.

DATA

The basis of the analysis is a dump of the European Open Street Maps data, retrieved from the Geofabrik server [22].

The candidate sites selected are rest areas along the German motorways. By applying the filter `highway = services` and `highway = rest_area`, 1732 rest areas within Germany were identified. Of those, 1264 were located within a reasonable $5\,\mathrm{km}$ margin from the TEN-T-network.

In order to compute the road network coverage, the road network is first extracted from the Open Street Maps (OSM) data as a routable graph using the open-source software osm2po [23]. For the sake of simplicity and performance, minor roads such as residential roads were neglected. The cost of each edge is equivalent to the time needed to complete it at the maximum allowable speed, the parameters are selected according to [18, Table A1]. As only spatial coverage is of interest, and minor links can be built relatively easy, reverse travel on links is allowed. This removed the influence of directionality when assessing the spatial coverage of a specific candidate site. To avoid optimization infeasibility, 353 of 48 329 road segments ($0.7\,\%$) were removed from the analysis, as they are further than $25\,\mathrm{km}$ away from the next rest area.

The distance matrix $\underline{\underline{\mathbf{A}}}$ is obtained by computing the shortest driving distance from each individual candidate site to all motorway road segments using `pgr_drivingDistance` function of the pgRouting library [24]. This is visualized in Figure 2: As the driving distance to a specific road segment is dependent on the road shape and routing, the iso-distance lines are not necessarily convex. Especially in short distances as relevant to the AFIR, the shape differs strongly from a circle.

The distance matrix $\underline{\underline{\mathbf{D}}}$ among candidate sites consists of the airline distances among all candidate sites. $\underline{\underline{\mathbf{Q}}}$ is thus diagonal and the main diagonal entries are zero. Figure 2 visualizes the process.

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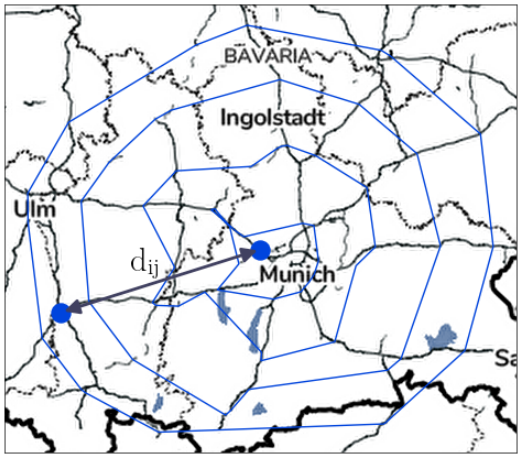


Fig. 2. The iso-distance lines around an exemplary candidate site (service area "Autohof Bergkirchen") extend along the major highways. Distances of 30, 60, 90 and 120 km are displayed as concave hulls. To comply with the AFIR regulation along TEN-T core routes, the next charging site has to be located within the 60 km ring around this site. For the objective function matrix $\underline{\underline{\mathbf{Q}}}$, the airline distances d_{ij} are used. Credit map: OSM Contributors

The TEN-T initiative is an EU strategy designed to enhance the free movement of individuals and goods by creating a coherent, efficient and multimodal transport infrastructure network throughout the EU. The TEN-T road network is structured into two levels: the core network, targeted for completion by 2030, and the comprehensive network, by 2050. The TEN-T freeway network examined in this study reflects the status of the TEN-T regulation as of 2024 and is obtained from the TENtec Interactive Map Viewer [5]. The TENT-T core highway network used in this study covers a length of 6358 km in Germany and the comprehensive network covers a length of 4271 km.

To evaluate the utility of charging networks, truck traffic flows of the year 2019 as published in [21] are used to compute the optimal ramp-up strategies.

Simulation assumptions. The simulated vehicles have a usable battery capacity of 500 kWh and the minimum permissible SOC is at 15 %. The simulated payload (influencing the consumption) is 19.3 t and the permissible detour to a charging site is 10 km.

RESULTS

Tis methodology aims to design a minimal charging network for Germany based on the processed data, compliant with the EU AFIR Regulation [4] along the TEN-T road network. Figure 3 shows the result of the optimization Equation 5. The target requirements for 2030 can be achieved with 135 electrified service areas, equaling to 1 charging site per 78.7 km of TEN-T road, or 100.7 km of motorway in total. Junctions appear to be preferable locations, congruent with the intuitive assumption. These sites can provide coverage for multiple motorways simultaneously. Especially the junctions between core and comprehensive network are frequently electrified, as this often suffices to meet the lower coverage requirements. This leads to relatively large portions of the comprehensive network not being equipped

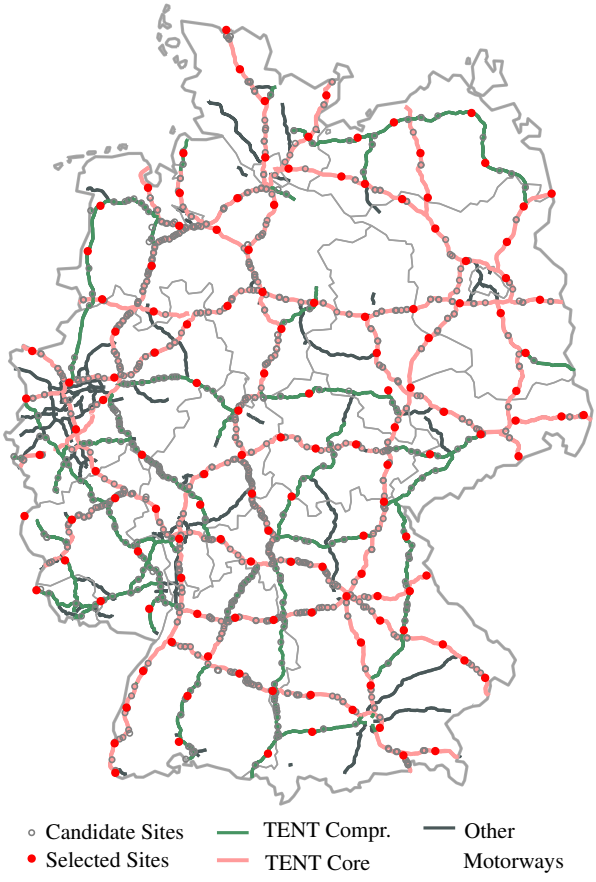


Fig. 3. Optimized AFIR-compliant charging network consisting of 135 out of 1732 candidate sites. Approx. 80 % of German motorways are associated with either the core network (red) or the comprehensive network (green), as Germany sits on the junction of 5 of the 9 TEN-T corridors [5].

with dedicated charging infrastructure apart from the sites shared with the core network, e.g. the A7 in southern central part of Germany. Along longer linear stretches of highway, additional charging sites are installed.

To compare this approach to previous research, the coverage constraint for the core network was tightened to 50 km instead of 60 km, equivalent to the close-meshed network approach in [11, p. 9]. We found a minimal set of 171 charging sites, 36 % lower than the benchmark. The electrification of each site requires costly processes, such as an electrical grid connection, and negotiations with land owners. Compared to small sites of 3 to 4 charging points (median, [11, p. 10]) this aggregation can bring economical benefits.

The charging network in Figure 3 represents a possible expansion stage for 2030 in Germany but does not provide a strategy for the successive installation of charging points. For this purpose, this paper analyzes two ramp-up strategies: Figure 4 compares the greedy strategy to 10 sampled random strategies. The key performance indicator is the share of the traffic flow that can successfully reach its destination. For that purpose, only long-haul traffic flow from [21] with distances of over 320 km was analyzed, in accordance with the filtering employed in [18]. The feasibility is dependent on the initial SOC of the vehicles, which corresponds to

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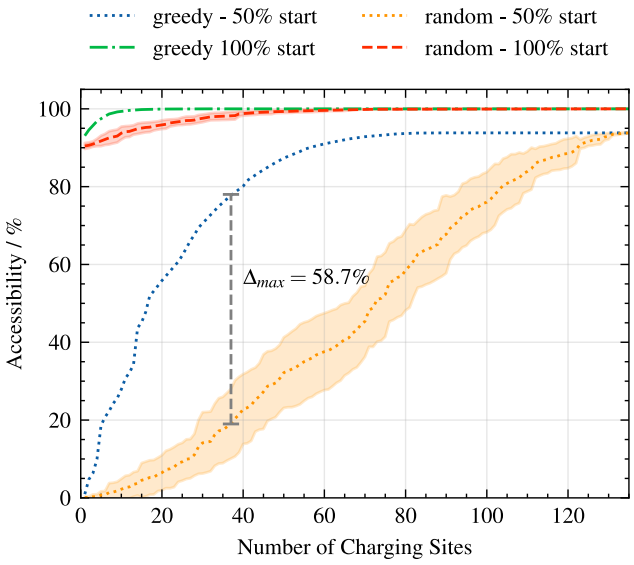


Fig. 4. Comparison of different ramp-up strategies for the optimized charging network from Figure 3. The initial SOC of the vehicle is either 50 % or 100 %. For each scenario, 10 runs of random selection are carried out. Mean and standard deviation of the results are visualized as lines and bands. The greedy approach iteratively picks the single candidate site, that provides the greatest increase in coverage of traffic flows.

available private charging infrastructure. To model different scenarios, the initial SOC was varied, too (50 % vs. 100 %).

Obviously, all strategies offer better coverage with a higher number of electrified sites. Yet the intermediate results vary strongly: Compared to an average random strategy at 50 % initial SOC, the greedy strategy can provide spatial coverage for up to 78 % of traffic flow, while the random approach only covers 19 % at 37 electrified sites, the greatest absolute difference found. Both strategies converge at 93.8 % coverage. This corresponds to the widely-known Pareto principle: 83 % of the final result can be achieved at only 27 % of the final network completed. The convergence is reached at 89 charging sites already. It should be noted, that further densification of the network beyond this point is not futile, as it would still improve spatial coverage, if the initial SOC is below 50 %. Furthermore, the time loss can still be significantly reduced for long-haul travel, as drivers can synchronize their mandatory breaks better with the charging processes of the vehicle [18].

At 100 % initial SOC, coverage is distinctly higher even without any public charging infrastructure, and the final spatial coverage is at 100 %. This can be explained, as many routes can be covered with the approx. 425 km range without recharging at all. Yet also here, the greedy strategy is dominant compared to random approaches in terms of spatial coverage.

In both scenarios, the greedy strategy performs better, than the range of the standard deviation of random approaches, even 2σ. It is thus extremely unlikely, that in an uncoordinated ramp-up such high coverages can be reached early on. Thus, we will finally explore the underlying approach of the greedy strategy to derive practical guidelines, by focusing on Figure 5.

The Figure 5 compares the different phases within the greedy strategy approach: The first sites electrified are geographically central and an already form a small but dense network. In the subsequent phase, the network is expanded radially. At 50 sites, almost full coverage is reached. The next 30 charging sites lead up to the convergence by enabling peripheral traffic flows to be realized with BET, and densifying the existing network in places. The final stage includes border crossings and unattractive sites. As only domestic traffic is modeled, border regions are naturally less attractive sites. Yet we do expect truly transeuropean long-haul BET traffic to pick up after domestic electric traffic, as it requires a coherent charging network throughout the EU. The greedy approach thus enables early applications for battery-electric transport, while also paving the way to an AFIR compliant comprehensive charging network throughout Europe.

Considering the optimality of the solution, it has to be noted that the greedy approach only increments the network by one station in each iteration. This could lead to attractive routes requiring more than one station to be covered only late in the process. The approach is thus not mathematically optimal, but the margin for optimization from an engineering

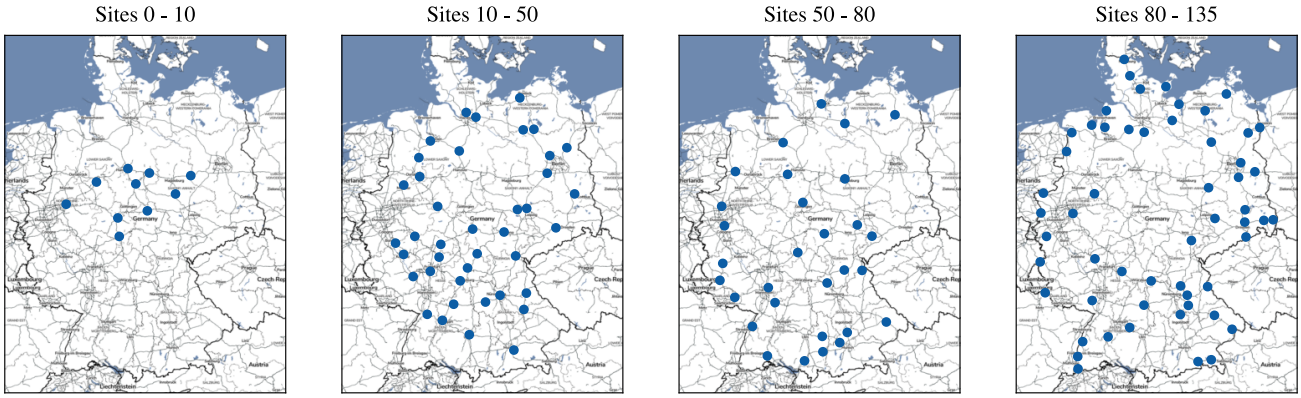


Fig. 5. Visualization of the greedy ramp up strategy at 50 % initial SOC. The first 10 charging sites are located centrally and relatively close to each other. A radial expansion follows. In the third stage, the density of the intermediate network is further increased. Finally, border regions are equipped with charging infrastructure. As only domestic traffic is evaluated, these contribute only marginal benefits to the system.

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perspective seems marginal, at the cost of strongly increased computation time. The current greedy approach model took $8\,\mathrm{h}$ of computation time on an Intel® Core™ i9-14900K CPU (Ubuntu 22.04 LTS).

CONCLUSION

This study presented a methodology for, first, the optimal spatial positioning of charging sites, to cover a real road network, and second an optimized ramp-up strategy for the strategic installation order of those sites. The aim is to provide early coverage of the freight flows for early adopters, while aiming for a fully covering and legally compliant target state.

Based on the traffic flow data and road network of Germany, the methodology yields 135 charging sites, which together ensure a maximum distance of $60\,\mathrm{km}$ between two sites along the TEN-T core network of Germany, as well as $100\,\mathrm{km}$ on the comprehensive network. Well suited locations often include core network junctions and core/comprehensive network junctions. Yet the current approach neglects the suitability of a site regarding the grid connection. This should be taken into account in future work, e.g. by incorporating distance of candidate site to the transmission lines into the optimization.

In comparing the random ramp-up processes with the greedy strategy, it could be found that the greedy strategy performs up to $58.7\,\%$ better. A spatial coverage of $80\,\%$ can be achieved by the strategic installation of just 40 charging sites in Germany. The strategy can generally be interpreted as an center-to-edge approach, which grows an initial charging network radially outwards. This raises the question, whether this pattern can be reproduced on a European level, which should be addressed in future research.

ABBREVIATIONS

The following notations are used in this manuscript: **AFIR** - Alternative Fuel Infrastructure Regulation, **BET** - Battery Electric Trucks , **NUTS** - Nomenclature of territorial units for statistics, **OSM** - Open Street Maps , **SOC** - State of Charge , **TEN-T** - Trans-European Transport Network,

$\underline{\mathbf{x}}$ - vector, $\underline{\underline{\mathbf{X}}}$ - matrix

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